FloatChat: ARGO Ocean Intelligence Platform

Enterprise Oceanographic Telemetry & AI Discovery System • Access Credentials Reference

Cloud Web Application: https://floatchat-argo-ocean-t7cjszmnhgb34rt6jnpgvz.streamlit.app/
GitHub Source: https://github.com/adityakittad1/floatchat-argo-ocean

Quick Access Instructions: Navigate to the live cloud URL above or run locally on port 8501. Enter either the **Username** or **Email Address** along with the designated password. FloatChat automatically verifies authentication and activates role-tailored feature sets, autonomous MCP SQL tools, and geospatial analytics.

1. PLATFORM AUTHENTICATION MATRIX

System Role	Login Username / Email	Password	Access Privileges & Capabilities
Administrator	**admin** or admin@argo.com	`admin123`	Full administrative authority, direct SQLite schema manager, NetCDF ingestion lab, and telemetry console.
Analyst	**analyst** or analyst@argo.com	`analyst123`	Interactive geospatial trajectory visualization, cycle analytics, CSV/Excel data export, and statistical summaries.
Researcher	**researcher** or researcher@argo.com	`research123`	AI Ocean Assistant powered by MCP SQL translation, deep CTD profile queries, and comparative depth models.
Scientist	**oceanographer** or oceanographer@argo.com	`ocean123`	Biogeochemical sensor diagnostic telemetry, 3D salinity/temperature profiles, and water mass stratification analysis.

2. TECHNICAL ARCHITECTURE & KEY MODULES

MCP Conversational SQL Engine: Translates natural language oceanographic inquiries into optimized SQL queries on argo.db.

Geospatial 3D Trajectory Map: Global interactive scatter maps tracking ARGO float deployment cycles, surface drift, and park depths.

Interactive CTD Visualizations: Dynamic Plotly graphs depicting Pressure vs Temperature & Salinity profiles with adaptive decimation.

NetCDF Automated Pipeline: Direct binary parser translating Argo .nc profile/tech files into structured relational database tables.

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